

Assessment brief: Network Science in the Wild

Your Task

Explore a real-world network of your choice — either by analysing its **structure**, its **dynamics**, or the **interaction between the two**. You may use an existing dataset or **construct your own**. Your goal is to pose and investigate a meaningful question using the methods we have studied in the module (you can also use methods you have learned through your readings of course).

You can approach the project through one or more of the following lenses:

- **Structure-focused:** What does the network’s structure reveal about the system? what are possible implications over its resilience?
- **Dynamics on network:** How do processes like diffusion, contagion, or coordination play out over the structure?
- **Dynamics of network:** How does the network evolve over time, and what shapes that evolution?
- **Dynamics as probe:** Can we use a process (e.g., percolation, spreading) to reveal ‘hidden’ structure?

What to Include

1. Your Network

- *Real-world* dataset (existing or created by you). It is perfectly acceptable to download a network from a repository (be it one discussed on the module or of your own choosing). Note that the fact that the primary network should be ‘real-world’ does not mean that you may not want to use synthetic networks as part of your analysis. So for example, you may want to generate random networks that preserve *some* properties in order to highlight the role of a particular network feature.

- If created by you: provide a short **Dataset Card** (as will be shortly discussed in the module). The marking scheme will recognise that creating/curating a dataset is, in itself, a valuable and potentially time-consuming Network Science contribution. For example, you may want to create your own dataset in order to answer a question for which there currently isn't any suitable dataset.

2. Your Question(s)

- What are you trying to understand? Why is it interesting?

3. Your Analysis

- Apply appropriate tools (e.g., degree distribution, k-core, clustering, spreading models, etc.)
- Use visualisations and statistics to support your interpretation

4. Your Interpretation

- Relate the findings to your question
- Reflect on insights, limitations, and assumptions

Submission

- **Format:** You may either submit a PDF with separate source code (since only file can be uploaded, generate a zip file containing both) or a Jupyter/Marimo Notebook (+ Dataset Card for people who created their own dataset – but see item *Network documentation* below for people who use an existing dataset). Please do make sure that your submission is self-contained. If your code is using non-standard libraries, make sure to either include them or to include instructions regarding how to download and install them. Document your code (i.e., comments!).
- **Network documentation:** I am asking people who create/curate their own dataset to submit a Dataset Card. This is because it follows from the so-called [FAIR guiding principles](#) of **F**indability, **A**ccessibility, **I**nteroperability, and **R**eusability. If you are using an existing dataset, you do not need to submit such a card. However, you **must** provide some description of the dataset you are using (as part of motivating why you picked that network for example). You should not assume that the reader will be familiar with it.
- **Length:** 3000 words. The number of words is indicative. Typically, it doesn't include bibliography and code (and dataset card in this case). Your report might be shorter or might be longer. If you are *significantly* shorter/longer than 3000 words, please get in touch for advice.
- **Deadline:** Please refer to Sussex Direct (the only authoritative source) but at the time of writing: 4pm on 18 May 2026 via Canvas.

Assessment Breakdown

Criterion	Weight
Quality of research question	15%
Soundness of structural/dynamic analysis, including dataset construction process if applicable	30%
Insight and interpretation	30%
Communication and clarity, including dataset documentation	25%

Need Inspiration?

- How does clustering shape diffusion of innovation in a tech ecosystem?
- Does the structure of a legislative network affect how laws get passed?
- Can you reverse-engineer the growth process behind a scientific collaboration network?
- What makes an influencer “structurally important” in meme propagation?